

S8-2

IE Error
(kcal mol⁻¹)IE Error
(kcal mol⁻¹)CPU Time
(log₁₀(s))Wall Time
(log₁₀(s))

HF	40.97	37.93	HF	wB97X-V	4.38	4.65
MP2	-8.64	-12.59	MP2	wB97X-D	4.22	4.57
PBE-D3	3.92	2.93	PBE-D3	B2PLYP-D3	4.03	4.4
B2PLYP-D3	1.43	-1.26	wB97X-V	MP2	3.99	4.39
wB97X-V	0.55	-0.76	B2PLYP-D3	PBE-D3	3.95	4.29
B3LYP-D3	0.29	-0.61	wB97X-D	B3LYP-D3	3.75	4.17
wB97X-D	-0.04	-0.07	B3LYP-D3	HF	3.65	4.14
	Pred.	Actual			Pred.	Actual

Da2

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wB97X-V	HF	37.6	36.29	HF	wB97X-V	4.35	4.61	wB97X-V
wB97X-D	MP2	-7.74	-14.05	MP2	wB97X-D	4.16	4.57	wB97X-D
B2PLYP-D3	PBE-D3	1.74	-3.07	wB97X-V	B2PLYP-D3	3.94	4.32	MP2
MP2	wB97X-D	-1.69	2.81	PBE-D3	B3LYP-D3	3.93	4.29	B2PLYP-D3
B3LYP-D3	B2PLYP-D3	1.17	-1.39	wB97X-V	MP2	3.9	4.23	B3LYP-D3
HF	B3LYP-D3	0.75	-0.96	B2PLYP-D3	PBE-D3	3.72	4.1	HF
PBE-D3	wB97X-V	0.35	0.43	B3LYP-D3	HF	3.57	4.09	PBE-D3
		Pred.	Actual			Pred.	Actual	

CBH

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HF	19.55	18.34	HF	wB97X-V	4.47	4.58
PBE-D3	-4.29	-3.59	PBE-D3	wB97X-D	4.25	4.44
wB97X-D	-3.08	-2.99	B3LYP-D3	B2PLYP-D3	4.08	4.23
B3LYP-D3	-2.17	-2.82	wB97X-V	MP2	4.02	4.1
B2PLYP-D3	0.67	-1.09	MP2	PBE-D3	3.97	4.09
wB97X-V	0.13	-0.54	B2PLYP-D3	HF	3.82	3.93
MP2	-0.08	-0.14			3.63	3.74
	Pred.	Actual			Pred.	Actual

C3A

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wB97X-V	HF	22.94	25.63	HF	wB97X-V	4.5	4.75	wB97X-V
wB97X-D	MP2	-6.06	-10.3	MP2	wB97X-D	4.35	4.68	wB97X-D
B2PLYP-D3	PBE-D3	3.24	2.51	PBE-D3	B2PLYP-D3	4.18	4.58	MP2
B3LYP-D3	wB97X-D	-0.94	-1.64	wB97X-D	MP2	4.15	4.47	B2PLYP-D3
MP2	B2PLYP-D3	0.47	-0.82	B2PLYP-D3	PBE-D3	4.06	4.38	PBE-D3
PBE-D3	B3LYP-D3	0.37	-0.64	wB97X-V	B3LYP-D3	3.85	4.33	B3LYP-D3
HF	wB97X-V	0.19	0.23	B3LYP-D3	HF	3.79	4.2	HF
		Pred.	Actual			Pred.	Actual	

C2C2PD

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HF	36.24	36.63	HF	wB97X-V	4.17	4.5
MP2	-11.65	-16.86	MP2	wB97X-D	3.99	4.46
PBE-D3	3.68	4.09	PBE-D3	B2PLYP-D3	3.76	4.01
B3LYP-D3	2.13	-2.7	wB97X-V	MP2	3.74	3.95
B2PLYP-D3	1.22	1.19	B3LYP-D3	MP2	3.71	3.87
wB97X-D	-0.9	-0.55	wB97X-V	PBE-D3	3.57	3.68
wB97X-V	0.66	nan	B2PLYP-D3	HF	3.39	nan
	Pred.	Actual			Pred.	Actual

2a

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wB97X-D	HF	46.88	44.51	HF	wB97X-V	4.49	4.73	wB97X-V
wB97X-V	MP2	-12.75	-14.61	MP2	wB97X-D	4.33	4.68	wB97X-D
B3LYP-D3	PBE-D3	4.34	-4.82	wB97X-V	B2PLYP-D3	4.14	4.56	B3LYP-D3
MP2	wB97X-V	1.83	2.8	PBE-D3	MP2	4.1	4.5	MP2
PBE-D3	B2PLYP-D3	1.31	-2.7	wB97X-V	B3LYP-D3	4.06	4.48	B2PLYP-D3
HF	wB97X-D	-1.26	-1.67	B2PLYP-D3	PBE-D3	3.84	4.24	PBE-D3
B2PLYP-D3	B3LYP-D3	1.22	-1.02	B3LYP-D3	HF	3.76	4.22	HF
		Pred.	Actual			Pred.	Actual	